SUPPORTING DOCUMENTS & MATHEMATICAL PROOFS

Handwritten Signature Verification Using Linear Algebra (Undergraduate Mini-Project)

1. Mathematical Derivations & Cauchy-Schwarz Bound

Theorem (Cauchy-Schwarz): For vectors u, v in Euclidean space R^N, |<u, v>| ≤ ||u||_2 ||v||_2.
Proof: For any real parameter t, consider the non-negative quadratic form f(t) = ||u - t v||_2² ≥ 0.
Expanding by inner product bilinearity: f(t) = <u - t v, u - t v> = ||u||² - 2t <u, v> + t² ||v||² ≥ 0.
Since this quadratic polynomial is non-negative for all real t, its discriminant Δ must be non-positive:
Δ = (-2 <u, v>)² - 4(||v||²)(||u||²) ≤ 0 &implies; 4<u, v>² ≤ 4 ||u||² ||v||² &implies; |<u, v>| ≤ ||u|| ||v||. Q.E.D.
Consequence: -1 ≤ cos(θ) = <u, v> / (||u|| ||v||) ≤ 1.

2. Hand-Calculated Worked Numerical Benchmark

Given benchmark vectors A = (1, 2, 3) and B = (2, 3, 4) in R³:
• Dot Product: 1(2) + 2(3) + 3(4) = 2 + 6 + 12 = **20**
• Norm of A: √(1 + 4 + 9) = √14 ≈ **3.7417**
• Norm of B: √(4 + 9 + 16) = √29 ≈ **5.3852**
• Euclidean Distance: √((-1)² + (-1)² + (-1)²) = √3 ≈ **1.7321**
• Cosine Similarity: 20 / (√14 × √29) = 20 / √406 ≈ **0.9926** (θ ≈ 6.98°)

3. Syllabus Mapping Matrix (Units 1, 2, and 3)

Unit	Mathematical Concept	Project Implementation	Status
Unit 1	Real Vector Spaces M_{m,n}(R)	Standardized canonical signature image matrices	Directly Implemented
Unit 1	Vectors in R^N	Flattened 1D signature embeddings	Directly Implemented
Unit 1	Subspaces	Subspace of signatures with fixed canvas size	Directly Implemented
Unit 1	Linear Independence	Signatures from distinct authors as non-collinear vectors	Directly Implemented
Unit 1	Basis & Dimension	Canonical standard basis vectors e_i of R^(mn)	Directly Implemented
Unit 1	Span & Sums of Subspaces	Linear combinations of stroke features and synthetic variants	Theoretical & Empirical Discussion
Unit 1	Direct Sums & Quotient Spaces	Orthogonal decomposition into signal + residual noise	Theoretical Discussion

4. Automated Verification Test Evidence

pytest 9.1.1 Execution Summary: 35 tests executed across test_vector_math.py, test_image_processing.py, test_comparison.py, and test_experiments_and_csv.py. **100% of tests passed (35 passed, 0 failed, 0 warnings).**

Linear Algebra Mini Project — Educational Demonstration
Notice: This application is an educational prototype and is not validated for legal, forensic, banking, or production identity authentication.